

Sinkhorn-HTI: an entropic substitute for HTI’s inner-loop c -transform

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Abstract

Hyperparameter Trajectory Inference (HTI; Amad and van der Schaar, 2026a) fits a conditional Lagrangian-OT surrogate by alternating between a potential / transport / spline update and an outer metric update. The potential update requires a per-batch Kantorovich c -transform whose hard form is a non-convex minimization over y_1 , which Amad and van der Schaar (2026a) solve with an inner L-BFGS loop. The first-order condition of the entropic dual is closed form: the soft c -transform, which replaces the inner loop by a single LogSumExp over the mini-batch; the transport-map regression target becomes the entropic barycentric projection of Pooladian and Niles-Weed (2021). On the semicircles benchmark of Amad and van der Schaar (2026a) at paper budget, the entropic variant ($\varepsilon = 3 \times 10^{-3}$) reaches held-out NLL = 12.05 and circle distance CD = 0.0130 (mean deviation from the unit circle) against 6.98 and 0.0158 for L-BFGS, and trains 3.3× faster. At fixed $\varepsilon > 0$ the entropic OT value carries an $\mathcal{O}(\varepsilon)$ bias (Pal, 2024).

1 Introduction

HTI (Amad and van der Schaar, 2026a) learns how a neural network’s conditional output distribution $p_{\theta_\lambda}(y \mid x)$ evolves with a continuous hyperparameter λ , given a sparse set of observed marginals. The method (Neural CLOT) extends the neural Lagrangian optimal-transport framework of Pooladian et al. (2024) to the conditional setting, and is trained by the alternating procedure of Algorithm 1 in Amad and van der Schaar (2026a). Inside this alternation, the Kantorovich potential g is updated against a semi-dual objective whose c -transform requires, for each source sample y_0 in a mini-batch, solving an inner minimization over candidate target points $y_1 \in \mathbb{R}^d$. Amad and van der Schaar (2026a) solve this via a short L-BFGS run warm-started from the transport-map network T (10 iterations for semicircles, cf Amad and van der Schaar 2026a).

Entropic OT yields a closed-form soft c -transform that replaces this inner minimization, at the cost of a bias controlled by a regularization strength ε . On the semicircles benchmark we implement both variants on top of the same network code and compare them at paper budget (seed 5, $N_{\text{outer}} = 2001$).

2 Background

For probability measures $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ with finite second moments and a continuous nonnegative cost c , the Kantorovich semi-dual (Villani, 2008)

$$\text{OT}_c(\mu, \nu) = \sup_{g \in L^1(\nu)} \int g^c d\mu + \int g d\nu, \quad g^c(y_0) := \inf_{y_1 \in \mathbb{R}^d} \{c(y_0, y_1) - g(y_1)\} \quad (1)$$

holds with one inner minimization per source y_0 in a single potential g . Adding a KL penalty against $\mu \otimes \nu$ to the primal,

$$\text{OT}_c^\varepsilon(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \int c \, d\pi + \varepsilon \text{KL}(\pi \parallel \mu \otimes \nu), \quad (2)$$

yields a strongly concave dual whose first-order condition in f at fixed g has a closed form (Peyré and Cuturi, 2019), the *soft c -transform*

$$g^{c, \varepsilon}(y_0) := -\varepsilon \log \int \exp\left(\frac{g(y_1) - c(y_0, y_1)}{\varepsilon}\right) d\nu(y_1), \quad (3)$$

with $g^{c, \varepsilon} \rightarrow g^c$ as $\varepsilon \rightarrow 0$ and an $\mathcal{O}(\varepsilon)$ bias on the OT value (Pal, 2024). HTI applies eq. (1) to conditional measures $\mu_k(\cdot \mid x)$ and $\mu_{k+1}(\cdot \mid x)$ at consecutive anchor times $t_0 < \dots < t_T$, with cost $c(y_0, y_1 \mid x)$ defined as the least action of a learned conditional Lagrangian $\mathcal{L}(q, \dot{q} \mid x) = \frac{1}{2} \dot{q}^\top G(q \mid x) \dot{q} - U(q \mid x)$ along paths q joining y_0 to y_1 .

For each source y_0 in a mini-batch, the resulting inner minimization over y_1 is solved by a short L-BFGS run warm-started from a transport-map network T (Algorithm 1, line 10 of Amad and van der Schaar 2026a). We keep the density-bias potential U , the learned metric G , and the spline head S (a small MLP that proposes the candidate path q used in the action integral) exactly as in Amad and van der Schaar (2026a).

3 The substitution

Derivation. Starting from the entropic dual of eq. (2),

$$\text{OT}_c^\varepsilon(\mu, \nu) = \sup_{f, g} \left\{ \int f \, d\mu + \int g \, d\nu - \varepsilon \iint \exp\left(\frac{f(y_0) + g(y_1) - c(y_0, y_1)}{\varepsilon}\right) d\mu(y_0) d\nu(y_1) + \varepsilon \right\},$$

the objective is strictly concave in f at fixed g for $\varepsilon > 0$ (Peyré and Cuturi, 2019), and the first-order condition in f gives $f^* = g^{c, \varepsilon}$. Replacing ν in eq. (3) by its mini-batch empirical version yields the empirical soft c -transform

$$\hat{g}_B^{c, \varepsilon}(y_k^{(i)} \mid x) = -\varepsilon \log \sum_{j=1}^B \exp\left[\frac{g_{\theta_g, k}(y_{k+1}^{(j)} \mid x) - c(y_k^{(i)}, y_{k+1}^{(j)} \mid x)}{\varepsilon}\right]. \quad (4)$$

The standard $1/B$ -normalized form differs by a constant $\varepsilon \log B$ that is y_k - and θ_g -independent and so vanishes in the dual-loss gradient.

Map-regression target. Replacing line 10 of Algorithm 1 in Amad and van der Schaar (2026a) by eq. (4) also changes the regression target for the transport-map network. HTI’s baseline regresses $T_{\theta_T, k}$ against the inner L-BFGS argmin; the natural entropic counterpart is the barycentric projection (Pooladian and Niles-Weed, 2021), i.e., the conditional expectation of the target under the same soft entropic plan, with the same $\exp((g - c)/\varepsilon)$ weights as eq. (4):

$$\hat{T}^\varepsilon(y_k^{(i)} \mid x) = \sum_{j=1}^B w_{ij} y_{k+1}^{(j)}, \quad w_{ij} \propto \exp\left(\frac{g_{\theta_g, k}(y_{k+1}^{(j)} \mid x) - c(y_k^{(i)}, y_{k+1}^{(j)} \mid x)}{\varepsilon}\right). \quad (5)$$

Equation (5) is the barycentric projection of the entropic OT plan (Pooladian and Niles-Weed, 2021) and recovers the hard argmin as $\varepsilon \rightarrow 0$. In HTI’s baseline, $T_{\theta_T, k}$ plays two coupled roles per outer step: warm-starting the inner L-BFGS run, and being regressed against its argmin. The entropic variant breaks this coupling: the barycentric projection in eq. (5) is computed from the cost matrix alone, independently of T . All other HTI components (metric, potential, spline head, outer update) are unchanged.

Remark. Equation (4) replaces the per-sample L-BFGS argmin by a batch-wise LogSumExp (the $-\varepsilon \log \sum_j \exp(\cdot)$ in eq. (4)). L-BFGS is tighter for well-converged g with a good T warm start.

4 Experiments

Framework and budget. We reimplement HTI in JAX/Equinox. Outer budget $N_{\text{outer}} = 2001$, inner budget $N_{\text{inner}} = 10$, and 10 L-BFGS iterations per source all match the choices made in Amad and van der Schaar (2026a). Both methods use mini-batch $B = 64$, which puts their per-step compute in the same range: the entropic step builds a single $B \times B$ cost matrix, while the L-BFGS variant runs B short inner optimizations of 10 iterations each. The data has four conditions, three anchor times $t \in \{0, 0.5, 1\}$, and 100 samples per condition.

Architectures and hyperparameters. Architectures match Table 8 of Amad and van der Schaar (2026a) with two deliberate changes: a smaller spline MLP ($[512, 512]$ vs $[1024, 1024]$) and an explicit eigenvalue floor $\lambda_{\min} = 0.3$ (the paper states none); other Lagrangian hyperparameters $(\alpha, h_y) = (0.05, 0.05)$ follow the original choices (α is the density-potential strength, h_y is the KDE bandwidth used inside it). The entropic regularization is fixed at $\varepsilon = 3 \times 10^{-3}$.

Outer metric optimizer. The values used in Amad and van der Schaar (2026a) and the released code (Amad and van der Schaar, 2026b) ($\eta_G = 5 \times 10^{-3}$, and gradient clip 1) produced a late-training metric-loss blow-up in our runs (see Section 5); we therefore use a constant $\eta_G = 5 \times 10^{-4}$ and tighten gradient-norm clipping to 0.25. The official HTI code (Amad and van der Schaar, 2026b) was only released after running the experiments and building the code; relative to its semicircles configuration we make two further changes: mini-batch $B = 64$ (release: 1024) and eigenvalue floor $\lambda_{\min} = 0.3$ (release: 0.1). The smaller B is forced by our limited compute budget and the entropic step’s $\mathcal{O}(B^2)$ memory cost.

Evaluation. Held-out metrics are computed at $t \in \{0.25, 0.75\}$: (i) NLL via a Gaussian-KDE of surrogate samples (bandwidth 0.05) evaluated at new ground-truth samples per condition, and (ii) the circle distance

$$\text{CD} := \mathbb{E}[\|y - \text{center}\|_2 - 1],$$

the mean deviation of generated points from the unit circle of the corresponding semicircle condition. Seed 5 was selected as the best-performing of five exploratory seeds at $N_{\text{outer}} = 400$.

Training dynamics. Figure 1 shows the four losses logged each outer iteration (potential, map, spline, and outer metric). Both methods drive their losses down over roughly the first 500 iterations and then plateau. The L-BFGS dual and metric panels carry occasional spikes: the inner problem is non-convex, so a metric step can shift which local minimum L-BFGS lands on, and the per-sample dual estimate jumps. The entropic dual is a LogSumExp and stays smooth. The two map losses are not directly comparable: our regression target depends on which target samples fall in the current mini-batch, whereas the L-BFGS argmin is batch-independent.

Held-out metrics. Table 1 reports wall-clock training time, the mean held-out NLL and CD over the two evaluation times, and the per-time breakdown. The headline NLL gap is concentrated at $t = 0.25$ (closer to anchor $t = 0$); the gap at $t = 0.75$ is narrower in NLL but reverses on CD.

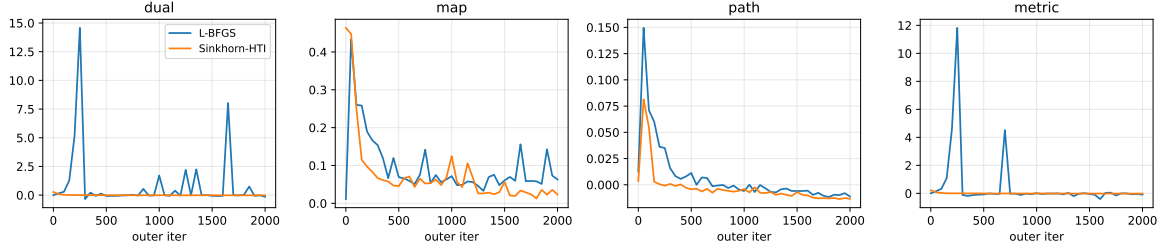


Figure 1: Training-loss trajectories. Panels: dual $\mathcal{L}_{\text{dual}}$, map $\mathcal{L}_{\text{map}}$, path $\mathcal{L}_{\text{path}}$, outer-metric $\mathcal{L}_{\text{metric}}$. Each outer iteration runs one N_{inner} -pass per anchor interval, giving $2 N_{\text{inner}}$ inner steps in total. Each plotted point is the mean over those step losses.

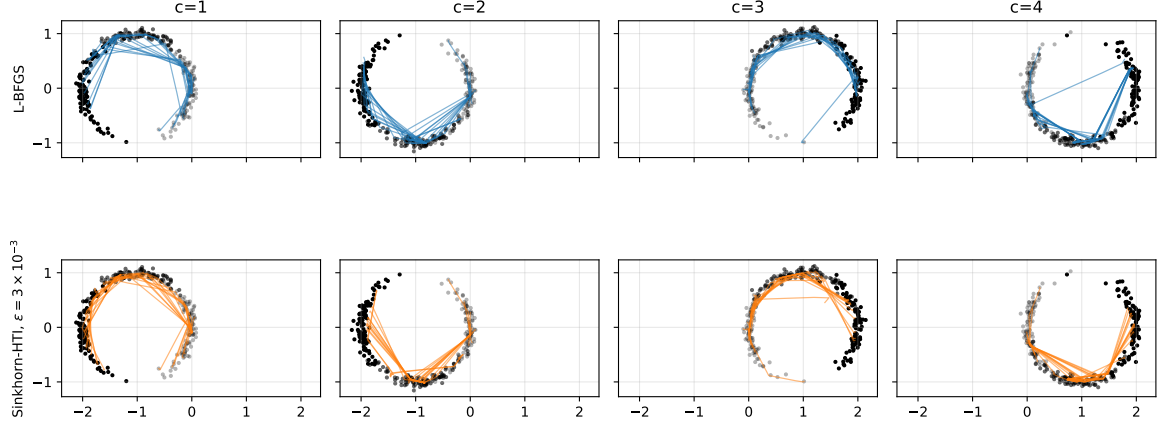


Figure 2: Surrogate trajectories. Dots: ground-truth samples at $t \in \{0, 0.25, 0.5, 0.75, 1\}$ with increasing opacity. Lines: surrogate paths from composing $T_{\theta_T, k}$ and the spline head.

Method	Time (s) ↓	Mean		$t = 0.25$		$t = 0.75$	
		NLL ↓	CD ↓	NLL ↓	CD ↓	NLL ↓	CD ↓
L-BFGS inner solver	12461	6.98	0.0158	10.13	0.0119	3.83	0.0198
Sinkhorn-HTI, $\varepsilon = 3 \times 10^{-3}$	3808	12.05	0.0130	19.72	0.0051	4.39	0.0210

Table 1: Semicircles held-out metrics. Mean is the average over $t \in \{0.25, 0.75\}$; wall-clock ratio $12461/3808 \approx 3.3\times$.

Qualitative reconstruction. Figure 2 plots surrogate-predicted trajectories for 12 starting samples per condition alongside the ground-truth process. Both methods recover the circular geometry.

5 Discussion

Headline. The two methods share architecture, initialization, data, anchor RNG, batch size, and outer-optimizer schedule, so the gap in Table 1 is attributable to the inner-solver substitution. L-BFGS attains lower KDE-NLL; the entropic variant attains lower CD and trains roughly $3.3\times$ faster.

Outer learning rate. The metric learning rate and gradient clip used in Amad and van der Schaar (2026a) and the released code (Amad and van der Schaar, 2026b), $\eta_G = 5 \times 10^{-3}$ and clip 1, made training unstable in our runs: the metric loss kept growing late on, and the

final spline drifted off the unit circles. We instead use a smaller $\eta_G = 5 \times 10^{-4}$ and a tighter clip of 0.25, which keeps training stable. The entropic variant runs with the same outer-loop settings, so the comparison only reflects the change in inner solver.

Inner-solver budget. We follow the choice in Amad and van der Schaar (2026a) with 10 L-BFGS iterations. Larger inner budgets produce a degenerate transport map with $T_{\theta_T, k}$ collapsing to the identity because the converged y_1^* from L-BFGS sits near its warm start $T(y_0)$, leaving the map MSE near zero throughout training. The entropic variant’s regression target is the barycentric projection, computed independently of T , and is unaffected.

Lagrangian-strength ablation. An exploratory run at $(\alpha, h_y) = (1.0, 0.1)$ substantially degraded held-out KDE-NLL on both methods: the strong- α regime over-pulls geodesics toward density ridges, hurting likelihood at held-out times without however fully breaking CD. The paper’s $(\alpha, h_y) = (0.05, 0.05)$ is the right choice for our optimizer.

Computational cost. At matched $B = 64$, the entropic inner step is a single $\mathcal{O}(B^2)$ LogSumExp over the pre-built cost matrix, while the L-BFGS inner step is B independent per-sample runs of 10 iterations each. The L-BFGS variant therefore does roughly an order of magnitude more cost-function and gradient evaluations per mini-batch; the measured wall-clock ratio is $3.3\times$ (Table 1).

Regularization bias. At fixed $\varepsilon > 0$ the entropic OT value carries an $\mathcal{O}(\varepsilon)$ bias under standard regularity (Pal, 2024). The bias on the gradient that drives the outer metric update is harder to characterize, so we just pick $\varepsilon = 3 \times 10^{-3}$ and check the choice empirically. A sweep of the soft-dual on the Sinkhorn-HTI components (Figure 3) confirms that the dual is essentially flat for $\varepsilon \leq 10^{-2}$ and drops sharply for $\varepsilon \geq 10^{-1}$, so the training ε is well within the flat regime.

Independence. The entire framework was reimplemented from scratch using only the Amad and van der Schaar (2026a) paper (with help from an LLM); the official release (Amad and van der Schaar, 2026b) appeared after the experiments were done and was consulted only for a few values the paper does not list (gradient clip, default B , eigenvalue floor $\lambda_{\min}$). The soft c -transform substitution itself has no counterpart in the released code.

Scope. HTI’s separate limitations (multi-hyperparameter extension, chaotic dynamics, metric-bias correction) are not addressed by the inner-loop choice.

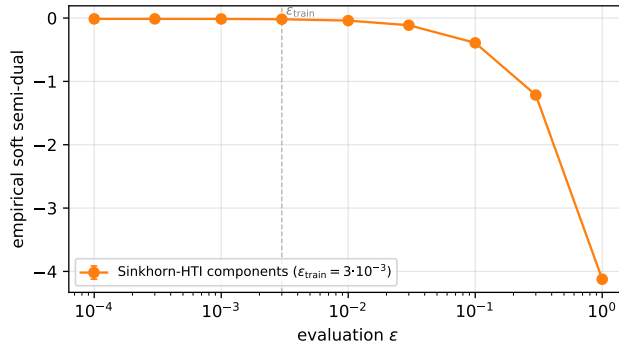


Figure 3: Empirical soft semi-dual on the Sinkhorn-HTI components at varying evaluation ε . Mean over 8 mini-batches. Vertical line: training $\varepsilon = 3 \times 10^{-3}$.

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